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Big five, linguistic styles, and successful online learning

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ABSTRACT

With the tremendous growth in online classes and programs at institutions of higher education, conflicting claims are increasingly being made on media and blog sites about the personality types that are compatible with this new learning format. The relations between the Five Factor Model of Personality (FFM) and linguistic styles with measures of academic performance in fully online asynchronous classes were examined. Consistent with findings based on research in face-to-face classes, Conscientiousness, Openness to Experience, and Analytic Thinking were associated with academic success. There was no support for the views propagated on the internet that extraversion, introversion, or anxiety hinder or foster online learning. The most robust predictor of academic performance throughout the semester, independent of the personality variables, was the number of words students contributed to the online discussions. These finding are interpreted within the Community of Inquiry framework (CoI; Garrison, 2003) and future lines of research are suggested.

1. Introduction

Over the past two decades, there has been a tremendous growth in online education at colleges and universities across the country (Allen & Seaman, 2017). In spite of the fact that a large body of research has demonstrated that student learning outcomes in online courses are comparable to traditional face-to face courses (for meta-analyses see Bernard et al., 2004; Means, Toyama, Murphy, Bakia, & Jones, 2013; Zhao, Lei, Yan, Lai, & Tan, 2005), surveys reveal that a majority of faculty continue to hold skeptical attitudes towards the newer teaching modality, with many expressing the view that online classes are less rigorous and effective than traditional classes in engaging students (e.g., Lederman, 2018; Shea, Bidjerano, & Vickers, 2016). Furthermore, in spite of the fact that the Babson Survey Research Group recently concluded that, with over a third of students at institutions of higher education now enrolled in online classes, this learning format has become "mainstream", various media as well as blog sites continue to propagate the view that it is likely to be uniquely suited to certain personality types (e.g., Top Ten, 2019, Chisum, 2018, Guay, 2017). To give an example, the U.S. News and World Report, which is the recognized leader in college rankings recently published a series of articles discussing the pros and cons of different types of online learning formats for introverts vs. extraverts. According to these articles, introverts tend to be more reflective and, therefore, would benefit from online programs that are asynchronous, self-paced, and do not require large amounts of group work. By contrast extraverts and ambiverts are likely to feel socially isolated in a totally online learning environment and should, therefore, select blended courses with a mix of live chats and face-to-face learning (e.g., Friedman, 2016; Fuster, 2017). Many colleges and universities are also increasingly offering prospective online students advice and feedback based on their scores on online readiness surveys (Wladis & Samuels, 2016). These surveys are widely used by institutions of higher education for screening purposes and typically contain a mixture of questions that assess attitudes towards the internet and technology, learning strategies, as well as psychological characteristics such as motivation, self-discipline, and assertiveness that are thought to be associated with positive online learning outcomes.

While the various recommendations offered to students by media and blogs sites as well as institutions of higher education sound plausible and are no doubt well-intentioned, the problem is that they are not based on empirical research. Typically, media and blogs sites draw on popular psychological literature or anecdotal observations to back up their various claims, which are often conflicting in nature. With respect to online readiness surveys, a recent large scale study found that no aspect of such surveys *differentially* predicts academic performance in online versus face-to-face courses, but students who received lower readiness scores were less likely to enroll in online classes the subsequent semester, suggesting the surveys may discourage some students from enrolling in such classes even if they are not at a higher risk (Wladis & Samuels, 2016).

One of the factors that is likely to have contributed to the

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dissemination of misleading information and unsubstantiated claims about online learning is that, in spite of repeated calls for research on student characteristics that are associated with successful online learning (e.g., Kauffman, 2015; Kerr, Rynearson, & Kerr, 2006; Wang & Newlin, 2000; Zhao et al., 2005), there has been surprisingly little systematic research on this topic, especially compared to traditional face-to-face classes. Given that the major impetus for the rapid growth in online education has been the convenience and flexibility this learning format offers students, the current trend is likely to continue for the foreseeable future, thereby increasing accessibility of higher education to many students as well as radically transforming it. It is, therefore, incumbent on researchers to provide students, faculty, and administrators with accurate and practical information regarding how the different modes of course delivery might impact the academic performance and experiences of students.

This study seeks to shed further light on student characteristics associated with successful online learning by focusing specifically on the relations between the Five Factor Model (FFM) of personality and linguistic styles with academic performance. The rationale for focusing on these two sets of variables is that they have been found to be robust predictors of academic performance in traditional classroom courses; thus, a large data base exists to serve as a bench mark for interpreting the findings from this study. To date, research examining the relations between the FFM with academic performance and linguistics styles with academic performance have been conducted independently of one another. It is, therefore, unclear how the personality and linguistic variables are associated with one another as well as whether the two sets of variables make an independent contribution to predicting academic success. A subsidiary goal of this study is, therefore, to not only integrate the research on academic performance in traditional face to face classes with online learning, but to also integrate the research on FFM with linguistic styles.

1.1. Background

Given that traditional face-to-face courses have been around much longer than online courses, it is not surprising that there is more extensive research on the psychological characteristics associated with academic success in traditional classroom courses. In the literature reviews on the FFM and linguistic styles, a summary of the findings based on traditional face-to-face courses will, therefore, be presented first and then be followed by findings based on online courses.

1.1.1. Five Factor Model of Personality and academic performance

A large body of research has examined the relations between the FFM and academic performance in traditional classroom courses (for comprehensive meta-analyses and reviews see: Poropat, 2009; Richardson, Abraham, & Bond, 2012; Vedel, 2014). According to the FFM, most personality traits studied by psychologists and used by laypersons to describe themselves and others can be adequately captured by five broad dimensions: Extraversion, Agreeableness, Conscientiousness, Openness to Experience, and Neuroticism. Extraversion assesses sociability, enthusiasm, and pleasurable arousal. Agreeableness assesses kindness, cooperativeness, and considerateness. Conscientiousness assesses planfulness, persistence, and goal-directed behavior. Openness to Experience assesses curiosity, imagination, and creativity. Neuroticism assesses higher levels of anxiety and depressed mood as well as lower levels of self-esteem.

Of the five dimensions of the FFM, the personality trait that has shown the most consistent and robust associations with academic performance is Conscientiousness (e.g., Poropat, 2009, Richardson et al.,2012; Vedel, 2014). Quite impressively, these associations have been found across primary to tertiary levels of education, even after controlling for other relevant variables such as intelligence. While the exact mechanism whereby Conscientiousness is associated with academic performance is not fully understood, several possible explanations have been offered. Conscientiousness is strongly related to a dimension of temperament referred to as effortful control, which is involved in self-regulation, and that plays a central role in the ability to intentionally initiate and inhibit actions as well as to direct and sustain attention on tasks (Abe, 2005; Rothbart, 2007). In the educational context, conscientious students have been found to score high on achievement motivation (Richardson & Abraham, 2009) and to use self-regulatory learning strategies as well as adaptive coping strategies (e.g., Bidjerano & Dai, 2007; Saklofske, Austin, Mastoras, Beaton, & Osbourne, 2012).

Following Conscientiousness, the personality dimension that has been the most likely to be associated with academic performance at the tertiary level of education is Openness to Experience (Vedel & Poropat, 2017). Individuals who score high on this personality dimension are intrinsically motivated to seek out intellectually challenging activities as well as novel experiences which, in turn, are thought to contribute to cognitive growth and accumulation of knowledge. Openness thus belongs among the “intellectual investment traits” which is considered to be one of the three pillars of academic performance along with effort and ability (von Stumm, Hell, & Chamorro-Premuzic, 2011).

While Conscientiousness and Openness are both positively correlated with self-regulated learning strategies, they show different patterns of relations with learning processes and approaches. More specifically, whereas Conscientiousness is associated with all four learning processes (methodological study, fact retention, elaborative processing and synthesis-analysis), Openness is associated with only elaborative processing and synthesis-analysis (Komarraju, Karau, Schmeck, & Avic, 2011). Another study found that of the five factors of personality, only Openness to Experience was negatively correlated with surface learning approaches and positively correlated with deep learning approaches (Chamorro-Premuzic & Furnham, 2009). Thus, Openness to Experience is likely to be associated with academic performance on tasks involving deeper processing of information and creative synthesis of ideas.

Interestingly, while Agreeableness, Extraversion, and Neuroticism are associated with academic performance at the primary and to some extent at the secondary levels of education, they have generally shown negligible or at best modest associations with academic performance at the tertiary level of education. According to Vedel and Poropat (2017), these differential pattern of relations may be due to the fact that as students progress through educational levels and are expected to become increasingly independent learners, these three personality traits are associated with countervailing tendencies in relations to academic performance. For example, extraverted individuals are sociable and assertive and, therefore, may be more likely to seek out help from both their teachers and peers when they are experiencing academic difficulties. However, they are also predisposed to seek out socially stimulating activities, which may distract from time spent studying on their own.

In contrast to the large body of research examining the relations between personality characteristics with academic performance in traditional face-to-face classes, a much smaller body of research is available on the personality characteristics of students who do well in online classes. Two studies have used the FFM to examine positive perceptions of and satisfaction with the online learning experience (Cohen & Baruth, 2017; Keller & Karau, 2013). These studies found that Conscientious and Openness to Experience were associated with positive perceptions of and satisfaction with online learning experiences; unfortunately, the associations between the Big Five and academic performance were not examined.

Other lines of research on personality characteristics and online learning have focused on more narrow band traits or just one of the FFM dimensions. For example, early research on personality and online learning focused on psychological constructs such as locus of control and self-efficacy (Wang & Newlin, 2000, 2002). Locus of control refers to generalized expectancies about the extent to which individuals perceive events in their life as under their control. Self-efficacy refers to

generalized expectancies about how well one can perform across a variety of situations. Both constructs were hypothesized to be predictive of performance in online classes because this learning format presumably demands a high level of self-regulation and self-direction. In general, the research on online education has provided support for the view that an internal locus of control and a high level of self-efficacy are associated with positive learning outcomes in online courses (e.g., Gokcearsaln & Alper, 2015; Joo, Lim, & Kim, 2013; Tsai, Chuang, Liang, & Tsai, 2011; Wang & Newlin, 2000; Yukselturk & Bulut, 2007).

Another line of research examining associations between personality characteristics and online learning has focused on personality types such as introversion and extraversion. In a recent literature review on the role of personality in computer based learning, researchers identified sixteen published studies, most of which utilized the Myer Briggs Type Indicator (MBTI) (Tlili, Esssalmi, Jemni, Kinshuk, & Chen, 2016). The MBTI divides personality and ways of processing information into four dichotomies which, in turn, are used to categorize people into 16 personality types. The review revealed that past research has yielded highly conflicting findings with respect to the associations between personality types and both satisfaction with and performance in online classes. For example, a study based on upper level university students who had taken online classes found that introverted type students preferred online whereas extraverted types preferred face-to-face classes (Harrington & Loffredo, 2010); however, another study based on introductory psychology students reported that extraverted and introverted students did not differ in terms of overall levels of satisfaction or performance in an online class. Similarly, studies have yielded conflicting findings with respect to participation in the online discussions, with some studies reporting that extraverted students participate more, but others reporting that introverts may behave more like extraverts online (Downing & Chim, 2004).

In summary, there are major gaps in our understanding of the relations between personality characteristics and academic performance in online classes. Even though a large body of research has demonstrated that Conscientiousness and Openness are associated with academic performance in traditional classroom courses, a recent meta-analysis of self-regulated learning strategies revealed the need to exercise caution in extrapolating from findings based on research conducted with traditional face-to-face classes to online classes (Broadbent & Poon, 2015). Similarly, even though a few studies have found that Conscientiousness and Openness are associated with positive perceptions of and satisfaction with online classes, previous research on online learning has revealed that learner satisfaction does not necessarily translate to learner achievement (e.g., Joo et al., 2013). In spite of the fact that the associations between Extraversion-Introversion and online learning has received considerable attention on media and blog sites, the limited research available on this topic has yielded conflicting findings, probably largely due to the fact that most of the studies used MBTI, which has poor psychometric properties (e.g., Pittinger, 1993).

1.1.2. Linguistic styles and academic performance

Another promising line of research on student characteristics associated with academic success focuses on linguistic styles (e.g., Pennebaker, Chung, Frazee, Lavergne, & Beaver, 2014; Pennebaker & King, 1999). These studies have typically analyzed students' writing samples with the Linguistic Inquiry and Word Count (LIWC), which is a widely used and well-validated computerized text analysis program. In a large scale study, Pennebaker and colleagues (Pennebaker et al., 2014) analyzed the college admissions essays of 25,000 entering students using the LIWC and identified eight standard function word categories that predicted college grades over four years. Students who obtained higher GPAs used higher levels of articles and prepositions in their essays, which suggest they were making references to complexly organized objects and concepts and thus thinking logically and hierarchically. By contrast, students who received lower GPAs used lower levels of articles and prepositions and higher levels of pronouns, auxiliary verbs, adverbs, as well as other parts of speech, which indicate that they were engaging in an intuitive or narrative mode of thinking. Based on these findings, Pennebaker and colleagues developed an algorithm for assessing individual differences in cognitive styles, which they refer to as Analytic Thinking. Individuals who score high on this measure tend to think more logically, whereas those who score low tend to think more intuitively.

In support of the view that Analytic Thinking may capture relatively stable individual differences in cognitive styles, Pennebaker and colleagues (Jordan & Pennbaker, 2017) found some striking differences among the candidates for the 2016 US primaries and presidential election on this measure. They analyzed transcripts of the debates using the LIWC and found that Donald Trump consistently scored far lower on the measure of Analytic Thinking than Hillary Clinton as well as the other Republican candidates, which accords with the observation made by several linguists that his speech style is unique among politicians for being more intuitive rather than logical (Golshan, 2017).

A growing body of research has also examined associations between linguistic styles and online learning (e.g., Joksimovic, Gasevic, Kovanovic, Adesope, & Hatala, 2014; Kovanovic, Joksimovic, Waters, Gasevic, Kitto et al., 2016; Yoo & Kim, 2013; Zhu, Herring, & Bonk, 2019). Most of these studies are based on Garrison and colleagues' Community of Inquiry Model of online learning (CoI; Garrison, 2003; Garrison, Anderson, & Archer, 2000) and have utilized LIWC and/or Coh Metrix (Graesser, McNamara, Louwerse, & Cai, 2004) to analyze online class discussions. According to the CoI model, online learning occurs through the reciprocal interaction of three components: teaching presence, cognitive presence, and social presence. The component that is considered to be the most basic to success in higher education is cognitive presence, which is online discourse that triggers intellectual stimulation and leads participants to explore higher order thinking. Joksimovic and colleagues found that even though distinct patterns of language use were associated with the different phases of cognitive presence (Joksimovic et al., 2014), only the LIWC measure of word count was predictive of higher levels of cognitive presence (Joksimovic et al., 2014; Kovanovic et al., 2016). The study by Zhu and colleagues (Zhu et al., 2019) found that students' cognitive presence was associated with higher levels of analytic thinking.

Integrating research on personality characteristics with text-analysis is particularly important for expanding our understanding of successful online learning because as Kovanovic and colleagues (2016) observe, in the absence of face-to-face interactions, online discussions become the main component of the entire educational experience. In fact, Garrison et al. (2000) state that what sets asynchronous fully online classes apart from face-to-face classes is that the communications and interactions are entirely text-based, which has two important implications. The first is that the participants must attempt to recreate the social and epistemic processes in traditional classroom courses by cultivating the three types of presences mentioned above, entirely through online discourse. The second implication is that because text-based communications allow more time for reflection and thus tend to be more disciplined and rigorous compared to oral communications, which tend to be fast-paced and fleeting, online classes may be more conducive to fostering higher order thinking. Consistent with this view, both first-year and senior online learners reported more deep approaches to learning in their coursework, including higher order thinking, reflective learning, and integrative learning than classroom learners (NSSE, 2008). In this study, measures of word count and Analytic Thinking extracted from the online discussions will, therefore, also be used as markers of the quantity and quality of class participation.

1.2. Research questions

This study examines the associations between personality and linguistic characteristics with academic performance in fully online asynchronous undergraduate psychology classes. The predictor

variables in this study were students' scores on the FFM as well as measures of Analytic Thinking and Word Count, which were extracted by analyzing students' online discussions and final papers using the LIWC. The two criterion measures used in this study were the average of the eight online quizzes students completed over the course of the semester and their grades on the final paper.

1. Are the FFM and linguistic dimensions associated with the measures of academic performance? Conscientiousness and Analytic Thinking will be positively correlated with quiz average because performing well on quizzes requires methodological study, fact retention, and the ability to categorize information into meaningful units. Openness to Experience and Analytic Thinking will be positively correlated with grade on final paper because writing a thoughtful paper requires deeper processing of information as well as the ability to creatively synthesize ideas. The measures of word count are also expected to be positively correlated with the measures of academic performance.

2. Are the FFM dimensions associated with the two measures of linguistic styles? Do the two set of variables make an independent contribution to predicting the outcome measures? Conscientious students are diligent and thorough. Conscientiousness will, therefore, be positively correlated with the measures of word count. Both Openness and Analytic Thinking involve deeper processing of information and the ability to organize it into hierarchically meaningful categories; thus, these two measures will be positively correlated with one another. Word count is hypothesized to be the most robust predictor of academic performance in this study because online learning is entirely text-based and words are the medium through which the three presences are cultivated (Garrison et al., 2000).

3. Do students exhibit a lower level of Analytic Thinking in the online discussions than their final paper? Based on previous findings that students are more likely to engage in deeper and higher order thinking in their position papers rather than threaded discussions or chats (Shea & Bidjerano, 2009), this study hypothesizes that students will exhibit a lower level of Analytic Thinking in the online discussions than their final paper.

4. Are the number of words students post in the online discussions correlated with their level of Analytic Thinking? Based on previous reports that word count in online discussions are predictive of higher levels of cognitive presence (Joksimovic et al., 2014; Kovanovic et al., 2016), this study hypothesizes that word count in the discussions will be positively correlated with level of Analytic Thinking.

2. Method

2.1. Participants

The participants in this study were recruited from six sections of fully online asynchronous undergraduate classes in personality psychology at a public comprehensive university. Students enrolled in these classes were offered the option of earning a few extra credit points towards the percentage score of one of the quizzes by either participating in this research study or completing an alternative assignment. The majority of the students (77%) opted to participate in the research study yielding a final sample size of 92 students ($N = 92$). A comparison of those who elected to participate and those who did not revealed that the two groups did not significantly differ with respect to their quiz average. Only students for whom both the online discussions and final paper were available were included in the study.

2.2. Assessment methods

*The Big Five Inventory (BFI;*John, Donahue, & Kentle, 1991*)*. The BFI is a widely used measure of the FFM and yields measures of five broad dimensions of personality. It consists of 44 items and asks respondents to rate on a five-point scale ranging from Disagree Strongly to Agree Strongly the extent to which they endorse each of the statements. In the present study, the alpha reliabilities for the personality dimensions ranged from 0.84 for Extraversion to 0.74 for Agreeableness.

*Linguistic Analysis and Word Count (LIWC2015;*Pennebaker, Boyd, Jordan, & Blackburn, 2015*)*. The LIWC is a well-validated text analysis program, which has been used in a wide range of studies (for review see Tausczik & Pennebaker, 2010). The program searches for target words or word stems from an extensive dictionary, categorizes them into linguistic dimensions, and then converts the raw counts to percentages of total words. The most recent version of the LIWC includes four composite scores, which were based on algorithms derived from past research: Analytic Thinking, Clout, Authenticity, and Tone. As previously mentioned, Analytic Thinking assesses the level of formal, logical and hierarchical thinking in the narrative. Clout assesses the level of confidence and expertise the writer projects. Authenticity assesses the level of directness and self-disclosure in the narrative. Tone assesses positive emotionality and is computed by subtracting negative emotion words from positive emotion words.

Quiz Average. The course was divided into five segments and students were required to complete two timed quizzes by designated due dates during each of the first four segments. The quizzes were based on chapters from the textbook and consisted of 22–35 items ($M = 28$). The quiz average was computed by converting the eight quizzes to percentage scores and calculating the average. Students were allowed three attempts on the quizzes, but the order of presenting the items was randomized for each attempt and they were not provided feedback on which items they answered correctly or incorrectly.

Online discussions. In addition to the quizzes, students were required to contribute to online discussions during each of the five segments. The topic of the discussions for the first four segments were broadly related to the theoretical approach to personality psychology that they were being tested on in the quizzes. However, the discussions were based on supplementary material (journal articles, videos, and online exercises) and not on the chapters from the textbook on which the quizzes were based. Students were explicitly instructed not to post specific questions related to the quizzes in the main discussion thread while the quizzes were ongoing, though they were invited to ask questions about the course and course material in a separate discussion thread.

The online discussions were conducted using the WebCT Vista blog format because it is author-centered and was thought to be more conducive to reflective posts than the discussion board format, which is topic-centered. The discussion for each of the segments involved completing several assignments (e.g., sharing reactions to a video and journal article as well as an online exercise); students were free to post their reflections for each of the segments as one combined post or to post them separately by assignment. In order to receive the maximum number of points for each of the discussions, students were required to complete all components of the discussion as well as to comment on at least one other student's post. All of the posts for each student was downloaded and saved as individual files. Comments to other students' posts and general questions about the class were deleted and the files were cleaned. The final files for the students, which included their initial introduction to the class as well as the five discussions, were analyzed using LIWC.

Grade on Final paper. For their final paper, students were required to watch four TED videos on happiness/well-being and to write a paper consisting of a minimum of 4 pages, double-spaced. They were provided with a one-page rubric, which outlined in detail the expectations for the paper. Students were instructed the paper must include: a) a summary of the videos; b) reflections on how the videos relate to the course material covered during the semester; c) personal reactions to the videos. Student papers were graded by the instructor without any knowledge of the students' scores on the BFI or the LIWC dimensions, several years prior to the analyses of the data for the present study. The papers were submitted via the assignment drop box and downloaded in electronic form. After each of the files were cleaned, they were processed with LIWC. Students' grades on the final paper were converted to

Table 1
Descriptive statistics for FFM and linguistic dimensions.

		Mean	SD
FFM			
	Extraversion	26.77	6.03
	Agreeableness	36.93	5.01
	Conscientiousness	35.51	5.32
	Neuroticism	24.57	6.15
	Openness to Experience	35.23	6.30
LIWC			
	Word Count (Online discussions)	2104.22	1098.41
	Word Count (Final paper)	1825.90	585.23
	Analytic Thinking (Online discussions)	45.29	14.60
	Analytic Thinking (Final paper)	58.25	15.61

Note: The means and SDs for the Word Counts displayed in the table are based on untransformed values.

a 13-point scale (A+ = 13 and F = 1).

3. Results

Prior to conducting the data analyses, the data were subjected to graphical and statistical examination. One student missed several quizzes and had a quiz average that was more than four standard deviations below the mean and transforming the score did not reduce the skew. Thus, this participant was excluded from the study. An analysis of the data with and without the outlier revealed that deleting it did not affect the overall pattern of results. Two students had word counts that were over 4 standard deviations above the mean. Thus, the word counts were subjected to square root transformations, which reduced the skew and normalized the distributions. The means and standard deviations for the FFM and linguistic dimensions are shown in Table 1.

3.1. Intercorrelations among FFM, linguistic dimensions, and outcome measures

The intercorrelations among the FFM, linguistic dimensions, and outcome measures are shown in Table 2. Several notable patterns emerged from the intercorrelations among the measures of academic performance and linguistic dimensions. Students' quiz average and grade on the final paper were robustly correlated with one another, indicating that students who did well on the online quizzes throughout the semester also tended to write strong final papers. In addition, the two measures of word count were moderately correlated with another, indicating that in general students who tended to write more in the discussion forums also tended to write longer papers. Similarly, the two measures of Analytic Thinking were also strongly correlated with one another, revealing students who exhibited higher levels of Analytic Thinking in the online discussions were also more likely to engage in higher levels of Analytic Thinking when writing their final papers.

Several notable patterns of findings also emerged with respect to the relations between the FFM and linguistic dimensions with the measures of academic performance. Consistent with expectations, Conscientiousness, Analytic thinking, and the two measures of word count were positively correlated with both quiz average and grade on final paper, whereas Openness to Experience was positively correlated only with grade on final paper. Also in keeping with expectations, Conscientiousness was positively correlated with both measures of word count and Openness to Experience was positively correlated with both measures of Analytic Thinking.

The other three dimensions of the FFM showed theoretically coherent patterns of relations with each of the personality dimensions, but with the exception of Agreeableness, which showed a trend towards being positively correlated with grade on final paper, they were generally orthogonal to both measures of academic performance.

3.2. BFI facets

The FFM is based on a hierarchical model of personality and each global dimension subsumes more specific personality characteristics, referred to as facets. Following the procedure outlined by Soto and John (2009), the scores for the two facet scales for each of the BFI dimensions were computed and their associations with the outcome measures were also examined.

In general, the BFI facet scales yielded similar patterns of results as the global dimensions. Both facets of Conscientiousness (self-discipline and order) as well as Openness to Experience (aesthetics and ideas) showed similar patterns of relations with the outcome measures as the global dimensions. Furthermore, both facets of Neuroticism (anxiety and depression) were orthogonal to the outcome measures.

The only exceptions to the foregoing patterns of findings were the Activity facet of Extraversion, which assesses energy and enthusiasm, and the Altruism fact of Agreeableness which assesses considerateness and helpfulness. The Activity facet was positively correlated with grade on final paper ($r = 0.23, p < .05$) as well as total number of words in online discussions ($r = 0.24, p < .01$) and negatively correlated with Analytic Thinking in the online discussions ($r = -0.21, p < .05$). The Altruism facet showed a trend towards being positively correlated with grade on final paper ($r = 0.16, p < .07$) as well as the number of words posted in the online discussion forum ($r = 0.16, p < .07$). The Assertiveness facet of Extraversion and the Compliance facet of Agreeableness were orthogonal to the outcome measures.

Table 2
Intercorrelations between FFM and linguistic dimensions with measures of academic performance.

		1	2	3	4	5	6	7	8	9	10	11
1.	Quiz Average	_										
2.	Grade on Paper	.50**	_									
3.	Word Count (online discussions)	.52**	.48**	_								
4.	Word Count (final paper)	.30**	.62**	.37**	_							
5.	Analytic (online discussions)	.32**	.22*	.24**	.02	_						
6.	Analytic (final paper)	.28*	.27**	.22**	.10	.64**	_					
7.	Extraversion	−.07	.03	.04	.01	−.15+	−.01	_				
8.	Agreeableness	.13	.16+	.12	−.02	.02	−.08	.21*	_			
9.	Conscientiousness	.48**	.37**	.35**	.20*	.10	.02	−.09	.33*	_		
10.	Neuroticism	−.10	−.02	−.03	.00	−.06	.07	−.05	−.23*	−.15+	_	
11.	Openness to Experience	.13	.35**	.16+	.00	.18*	.26**	.19*	.23*	.06	−.03	_

one-tailed.
+ $p < .10$.
* $p < .05$.
** $p < .01$.

Table 3
Multiple regression analyses predicting quiz average and grade on final paper from personality and linguistic dimensions.

Criterion Variable	Predictor	β	R^2	Adjusted R^2	F
Model 1					
Quiz Average	Conscientiousness	.46***	0.28	0.26	16.12***
	Analytic Thinking	.22*			
Model 2					
Quiz Average	Conscientiousness	.34***	0.38	0.36	17.33***
	Analytic Thinking	.16+			
	Word Count	.36***			
Model 1					
Final Paper Grade	Conscientiousness	.35***	0.31	0.28	9.10***
	Openness	.26*			
	Activity	.15			
	Analytic Thinking	.17+			
Model 2					
Final Paper Grade	Conscientiousness	.24**	0.58	0.56	36.13***
	Openness	.35***			
	Word Count	.56***			

+ $p < .10$.
* $p < .05$.
** $p < .005$.
*** $p < .0005$.

3.3. Regression analyses

To examine the relative contribution of the personality and linguistic variables in predicting students' academic performance, a series of regression analyses were conducted (see Table 3). The criterion variable for the first two regression analyses was quiz average. As Model 1 shows, the analysis with Conscientiousness and Analytic Thinking as predictor variables accounted for 26% of the total variance with both variables accounting for significant unique variance. Given that word count was also robustly associated with quiz average, the analysis was repeated by entering this variable along with the first two predictors. As Model 2 shows, the analysis with the three predictors accounted for 36% of the total variance with both Conscientiousness and word count accounting for significant unique variance. However, the addition of word count resulted in the unique variance for Analytic Thinking becoming only marginally significant.

The criterion variable for the next two regression analyses was grade on final paper. As Model 1 shows, the analysis with Conscientiousness, Openness to Experience, Activity, and Analytic Thinking as predictor variables accounted for 28% of the total variance with only Conscientiousness and Openness to Experience, accounting for significant unique variance. Given that word count was also robustly associated with grade on final paper, the analysis was repeated by entering this variable along with Conscientiousness and Openness to Experience. As Model 2 shows, this regression equation accounted for 56% of the total variance with all three of the variables accounting for significant unique variance.

The foregoing regression analyses revealed that Word Count was the most robust predictor of the measures of academic performance in this study. Conscientiousness, Analytic Thinking, and the Activity facet of Extraversion were found to be to be positively correlated with Word Count for online discussions. To gain a better understanding of the relative contribution of the three personality variables in predicting Word Count, a follow-up multiple regression analysis was conducted. The analysis revealed that the model accounted or 19.4% of the total variance $F\ (3,\ 84) = 7.98$, $p < .0005$, with Conscientiousness, Analytic Thinking, and Activity each accounting for significant unique variance, βs = 0.29, 0.23, 0.27, respectively, $ps < 0.01$ to 0.05.

3.4. Additional analyses

The correlational analyses revealed that students' level of Analytic Thinking in the online discussions was strongly correlated with their level of Analytic Thinking in their final paper. However, the mean level of Analytic Thinking was also found to be substantially lower in the online discussions than the final paper. A paired sample *t*-test was conducted to examine whether the difference in level Analytic Thinking was statistically significant. The results were highly significant ($t = 9.62$, $p < .0005$, two-tailed).

Given that as predicted, students exhibited lower levels of Analytic Thinking in the online discussions than in their final papers, some post-hoc analyses were conducted to examine whether there were other linguistic differences between the online discussions and the final papers, by conducting paired samples *t*-tests with the other three LIWC composite scores. The analyses revealed that students scored significantly higher on Authenticity ($t = 18.54$, $p < .0005$, two-tailed) as well as significantly lower on Clout and Tone ($ts = 14.73$ and 18.23, $ps < 0.0005$, two-tailed) in their online discussion posts compared to their final papers. Furthermore, the correlational analyses revealed that the level of Analytic Thinking in the online discussion posts was associated in the expected direction with both the levels of both Authenticity and Clout ($rs = -0.38$ and 0.36, $ps < 0.001$, two-tailed), but was orthogonal to Tone ($r = 0.05$).

4. Discussion

Consistent with expectations, Conscientiousness, Analytic Thinking, and Openness to Experience were associated with successful online learning. More specifically, Conscientiousness and Analytic Thinking were positively correlated with students' quiz average as well as their grade on the final paper, whereas Openness to Experience was associated only with grade on the final paper. These findings are in keeping with the view that performing well on the multiple choice quizzes primarily requires methodological study, fact retention, as well as the ability to categorize information into meaningful units, whereas writing a thoughtful final paper requires deeper processing of information as well as the ability to creatively synthesize ideas.

In contrast to Conscientiousness and Openness, the other dimensions of the FFM were either orthogonal to the measures of academic performance or showed weak and inconsistent patterns of relations, which is similar to findings based on face-to-face classes. Of the five personality dimensions, Extraversion is the one that has received the most attention on media and blog sites and has been the focus of most contradictory claims. To further complicate matters, research on this personality dimension has yielded conflicting findings, which is probably largely due to the fact that the bulk of the studies used the MBTI, which has been found to have poor psychometric properties. In this study, the global Extraversion dimension was orthogonal to both measures of academic performance as well as the amount students wrote in the online class discussions. However, when the associations were examined at the facet level, this study revealed somewhat surprisingly that it was the Activity facet, which assess enthusiasm and energy, and not the Assertiveness facet, which assesses talkativeness, sociability, and assertiveness, that was positively correlated with the amount students wrote in the online discussions. The Activity facet was also negatively correlated with Analytic Thinking in the discussions and positively correlated with grade on final paper. Based on the findings from this study, there is little evidence that students who are talkative, sociable, and outgoing are at an advantage or disadvantage in a fully online asynchronous learning environment. There is also little evidence that introverts are at an advantage or disadvantage in such an environment.

The other personality dimension that has received considerable attention on media and blog sites is Neuroticism. A recent survey revealed that some students use online courses as one way to manage social anxieties related to attending courses in person (Online Education Trends Report, 2018). However, some educators have expressed concerns that because students in online classes are not required to meet at

a specified place and time on a regular basis, this learning format may be associated with reduced structure and increased ambiguity which, in turn, may exacerbate anxiety levels in some students (Keller & Karau, 2013). The findings from this study revealed, however, that Neuroticism was not associated with the outcome measures at either the global or facet level. Based on the findings from this study, there is little evidence that students who tend to score higher on depressed mood or anxiety are adversely affected by or benefit from a totally online and asynchronous course format.

To sum up, the findings involving the FFM have important practical implications for faculty, students, and administrators. As stated at the outset, in spite of the rapid growth in online education and the large body of research showing that online courses are comparable to face-to face classes in terms of student learning outcomes, many faculty still have biased views towards online education. This study demonstrates that the same student characteristics that are predictive of high levels of academic performance in traditional face-to-face classes, namely, conscientiousness, analytic thinking, and openness to experience are associated with successful online learning. These findings should be reassuring to faculty who are concerned that online courses are inherently less rigorous and effective at engaging students than traditional classroom courses. Furthermore, contrary to the view propagated at various media, blog, and even college and university websites, there is little evidence that personality traits such as Extraversion or Neuroticism foster or hinder online learning either in terms of academic performance or class participation.

While the personality and linguistic characteristics associated with successful online learning are remarkably consistent with predictions, this study also revealed that the most robust predictor of both outcome measures was word count. In the case of the final paper, these findings are in some respects to be expected, given that the minimum length for the paper was only four pages. It is, therefore, not surprising that students who wrote longer papers were more likely to do a thorough job summarizing the major findings reported in the videos, articulating connections with the course material, and elaborating on their personal reactions. However, the robust associations between the word count for the online discussions and quiz average raises some intriguing questions. As previously mentioned, the online discussions were based on supplementary materials (videos, articles, and personality tests), whereas the quizzes were based on textbook chapters. Thus, the amount students wrote in the discussions cannot be simply construed as an index of how much students rehearsed the content of the chapters.

The question that arises is what possible mechanism underlies the robust associations between word count and performance on the quizzes. The most plausible explanation that can be offered is that it is due to their shared variance with Conscientiousness. Thus, students who were diligent in studying for the quizzes were also more likely to take the time writing longer discussions. Conversely, students who waited until the last minute to complete the quizzes were also more likely to hastily post brief reflections at the last minute. In support of this view and consistent with predictions, Conscientiousness was robustly correlated with both measures of word count in this study. However, the regression analyses revealed that word count accounted for significant unique variance over and above Conscientiousness and was an equally robust predictor of quiz average.

To fully account for why the amount students contributed to the online discussions was such a robust predictor of academic performance throughout the semester, it may be necessary to draw on the CoI framework. As previously mentioned, according to this model, successful online learning requires the reciprocal interaction of three types of presences: cognitive, social, and teaching. Cognitive presence involves discourse that triggers intellectual engagement and moves participants to higher order thinking. Social presence involves online discourse that promotes positive interactions and group cohesion. Teaching Presence supports and enhances social and cognitive processes to foster positive educational outcomes. More recently, Garrison and Akyol (2015) expanded on the CoI framework to include a metacognition construct which assesses self-direction and motivation as well as co-regulation of learning.

One possible explanation that can be offered for why word count in the online discussions was a more powerful predictor of academic performance than the personality variables is that it tapped into the three types of presences, whereas Conscientiousness tapped mainly into the metacognition construct. In keeping with this view, as previously mentioned, Kovanovic and colleagues (2016) found that word count was the most powerful predictor of cognitive presence. Similarly, a previous study on experiential learning (Abe, 2011a) found that the number of words students wrote in their practicum journals over the course of the semester was strongly correlated with how intellectually stimulating they rated their practicum experiences to be. A fruitful avenue for extending research on successful online learning would, therefore, be to administer the CoI Survey (Arbaugh et al., 2008) in addition to the FFM. Such research would enable us to examine whether the measures of the other presences, in particular, cognitive and social, can account for additional significant variance in predicting both word count as well as the measures of academic performance over and above the personality variables.

Several notable patterns of findings involving Analytic Thinking also emerged from this study. Consistent with the view that this measure assesses relatively stable individual differences in cognitive styles, the measure of Analytic Thinking that was extracted from the online discussions and from the final papers were robustly positively correlated with one another. Furthermore, the two measures of Analytic Thinking were, as expected, positively correlated with Openness to Experience. Nevertheless, this study also revealed that the mean level of Analytic Thinking was significantly lower in the online discussions than the final paper, which shows that it is also influenced by the context (Jordan & Pennbaker, 2017). These findings are consistent with previous report that online students are more likely engage in deeper and higher order thinking in their position papers rather than threaded discussions or chats (Shea & Bidjerano, 2009) and are also in keeping with expectations. The final paper was submitted to the instructor via the assignment drop box and even though it was not as formal a writing assignment as a traditional term paper, it was more structured than the online discussions. By contrast, the discussions were posted in an online forum so that students could freely share their thoughts and feelings with other students and comment on each other's posts. Thus, the finding that the mean level of Analytic Thinking was lower in the discussions than the final paper suggests that students were not merely "posting mini-essays" (Dennen, 2008) in the online forum, but were spontaneously adopting a more informal narrative style, without even being instructed to do so.

While one of the major objectives of higher education is to foster Analytic Thinking (Pennebaker et al., 2014), online discussions posts that are written in a highly formal, logical, and detached style are likely to interfere with the ability of the participants to project their personal characteristics into the community of inquiry and thereby present themselves as real people–both of which are vital to fostering social presence (Garrison et al., 2000). Some support for this view came from additional findings from this study[1] that students exhibited significantly higher levels of Authenticity (directness and self-disclosure) and lower levels of Clout (confidence and expertise) in their discussion posts than in their final papers and that these two measures were also associated in the expected direction with Analytic Thinking.

As Garrison and Akyol (2015) recently observed: "…open

[1] At first glance, the finding that students scored higher on Tone (which assesses positive emotionality) in their final papers than the online discussions might seem somewhat counterintuitive. However, these findings are most likely attributable to the fact that the topic of the final paper was happiness/well-being, whereas the online discussions focused on a variety of topics.

communication and group cohesion are dependent upon establishing a secure and supportive learning climate (i.e., social presence). This, however, raises interesting questions as to the nature of social presence and the impact of purely interpersonal communication (versus purposeful scholarly discourse) on cognitive presence (p. 70)." As previously mentioned, of the three components of the CoI framework, cognitive presence is considered to be the most central to success in higher education. Another research topic that warrants further investigation is, therefore, whether there is an "optimal" level of Analytic Thinking in online discourse for fostering cognitive presence, without undermining social presence, and vice versa. This research question can be addressed in part by administering the CoI survey and examining the associations between students' ratings of cognitive and social presences in a given class with the mean levels of Analytic Thinking, Authenticity, and Clout in the online discussions for that class. Future studies might also incorporate other LIWC cognitive composite measures (e.g., Abe, 2011b; Abe, 2012) to shed further light on the reciprocal interaction between cognitive and social presences.

In recent years, there has also been increasing debate over the pros and cons of emphasizing "quality" vs. "quantity" in online discussions (Dennen, 2008; Ransdell, 2013). This study revealed that the Activity facet of Extraversion was both positively correlated with the amount students wrote in the online discussions and negatively correlated with their level of Analytic Thinking in the online discussions—which may be reflective of the fact that enthusiastic people generally have high energy levels, but also can be somewhat impulsive at times (Abe, 2005). Nevertheless, this study revealed that the measure of word count and Analytic Thinking extracted from the discussions were robustly positively correlated with one another[2], indicating that overall there does not seem to be a trade-off between "quality" and "quantity" in the online discussions. In fact, at least based on findings from this study, encouraging students to write more may actually foster higher levels of Analytic Thinking as well as improve their performance on the online quizzes. Thus, instructors of online courses should be careful not to impose too many rigid rules in online discussions and should encourage students to freely share their thoughts and feelings with their classmates. Similar recommendations were offered by Gavesic and colleagues in their study on strategies to promote cognitive presence in asynchronous online discussions (Gasevic, Adesope, Joksimovic, & Kovanovic, 2015). It should, nevertheless, be noted that the results of this study may have been due in part to the fact that the blog format was used for the discussions, instead of the discussion board format. As previously mentioned, the blog format is author-centered and, therefore, may be more conducive to reflective posts than the discussion board format, which is topic-centered. An important avenue for future research would be to examine whether these different discussion formats have an impact on the quality and quantity of discussion posts.

4.1. Conclusion

In summary, this study revealed the fruitfulness of integrating research on FFM and linguistic styles to enhance our understanding of successful online learning. While the findings from this study demonstrated that the personality and linguistic characteristics of students who do well in online classes are theoretically coherent and similar to those in traditional face-to-face classes, this study needs to replicated in online classes in other majors as well as at other institutions. A growing body of research is revealing that the personality characteristics associated with educational success may vary somewhat across different academic majors (e.g., Vedel & Poropat, 2017). Similarly, research comparing students in face-to-face and online classes have found learning outcomes may vary across different disciplines (Zhao et al., 2005). Another important avenue for extending this study is to incorporate the CoI Survey as well as other LIWC linguistic variables, so many of the questions that that arose from this study can be further investigated.

While some caution must be exercised in overgeneralizing based on the findings from a single study, some practical advice can be tentatively offered to both faculty and to students. To faculty, the major message is that there is little evidence that extraverted, introverted, or anxious students are adversely affected by or benefit from the online format and that students who do well in face-to-face classes are also likely to do well in online classes. To students, the main message is to not be swayed too much by information regarding online education on media and blog sites, and to base decisions about selecting method of course delivery on personal preferences and convenience.

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Declaration of Competing Interest

None.

References

Abe, J. A. (2005). The predictive validity of the five-factor model of personality with preschool age children: A nine-year follow-up study. *Journal of Research in Personality, 39*, 395–486.
Abe, J. A. (2011a). Positive emotions, emotional intelligence, and successful experiential learning. *Personality and Individual Differences, 51*, 817–822.
Abe, J. A. (2011b). Changes in Alan Greenspan's language use across the economic cycle. *Journal of Language and Social Psychology, 30*, 212–223.
Abe, J. A. (2012). Cognitive-affective styles associated with position on war. *Journal of Language and Social Psychology, 31*, 212–222.
Allen, E., & Seaman, J. (2017). Digital learning compass: Distance education enrollment report. Retrieved from https://onlinelearningsurvey.com/reports/digtiallearningcompassenrollment2017.pdf.
Arbaugh, J. B., Cleveland-Innes, M., Diaz, S. R., Garrison, D. R., Ice, P., Richardson, & Swan, K. P. (2008). Developing a community of inquiry instrument: Testing a measure of the community of Inquiry framework using a multi-institutional sample. *The Internet and Higher Education, 11*(3–4), 133–136.
Bernard, R., Abrami, P., Lou, Y., Borokhovski, E., Wade, A., Wozney, L., et al. (2004). How does distance education compare to classroom instruction? A meta-analysis of the empirical literature. *Review of Educational Research, 74*, 379–439.
Bidjerano, T., & Dai, D. Y. (2007). The relationship between the big-five model of personality and self-regulated learning strategies. *Learning and Individual Differences, 17*, 69–81.
Broadbent, J., & Poon, W. L. (2015). Self-regulated learning strategies and academic achievement in online higher education learning environments: A systematic review. *Internet and Higher Education, 27*, 1–13.
Chamorro-Premuzic, T., & Furnham, A. (2009). Mainly openness: The relationship between the big five personality traits and learning approaches. *Learning and Individual Differences, 19*, 524–529.
Chisum, R. (2018). *May*. SHSU Online Campus: Introversion and the online classroom. https://online.shsu.edu/news/2018/may/introversion-and-the-online-classroom.html.
Cohen, A., & Baruth, O. (2017). Personality, learning, and satisfaction in fully online academic courses. *Computers in Human Behavior, 72*, 1–12.
Dennen, V. P. (2008). Looking for evidence of learning: Assessment and analysis of methods for online discourse. *Computers in Human Behavior, 24*, 205–219.
Downing, K., & Chim, T. (2004). Reflectors as online extraverts? *Journal of Educational Studies, 30*, 265–276.
Friedman, J. (2016, January). Examine the benefits, drawbacks of online learning for introverts. *U.S. News and World Report*. Retrieved from: https://www.usnews.com/education/online-education/articles/2016/01/05/examine-the-benefits-drawbacks-of-online-learning-for-introverts.
Fuster, B. (2017, May). Pick an online program as an introvert, extravert. *U.S. News and World Report*. Retrieved from: https://www.usnews.com/education/online-learning-lessons/articles/2017-05-12/tips-for-introverts-extroverts-to-choose-an-online-program.
Garrison, D. R. (2003). *Cognitive presence for effective asynchronous online learning: The role of reflective inquiry, self-direction, and meta-cognition.* Elements of Quality Online Education: Practice and Direction. Retrieved from https://www.researchgate.net/

[2] It should be emphasized that word count and Analytic Thinking are not always positively correlated with one another. In fact, in this study, word count and Analytic Thinking extracted from students' final papers were not positively correlated with one another. Across three samples, measures of word count and Analytic Thinking extracted from narrative reactions to political events were also found to be orthogonal to one another.

publication/228585404.
Garrison, D. R., & Akyol, Z. (2015). Toward the development of a metacognition construct for communities of inquiry. *Internet and Higher Education, 24*, 66–71.
Garrison, D. R., Anderson, T., & Archer, W. (2000). Critical inquiry in a text-based environment: Computer conferencing in higher education. *Internet and Higher Education, 2*(2–3), 87–105.
Gasevic, D., Adesope, O., Joksimovic, S., & Kovanovic, V. (2015). Externally-facilitated regulation scaffolding and role assignment to develop cognitive presence in asynchronous online discussions. *Internet and Higher Education, 24*, 53–65.
Gokcearsaln, A., & Alper, A. (2015). The effect of locus of control on learners' sense of community and academic success in the context of online learning communities. *Internet and Higher Education, 27*, 64–73.
Golshan, T. (January 2017). Donald Trump's unique speaking style explained by linguists. Retrieved from https://www.vox.com/policy-and-politics/2017/1/11/14238274/trumps-speaking-style-press-conference-linguists-explain.
Graesser, A. C., McNamara, D. S., Louwerse, M. M., & Cai, Z. (2004). Coh-Metrix: Analysis of text on cohesion and language. *Behavior Research Methods, 36*(2), 193–202.
Guay, M. (2017, December). *Giving voice to the introverted student: Online vs. traditional school.* https://www.huffpost.com/entry/online-vs-traditional-school- introverted_b_4518426.
Harrington, R., & Loffredo, D. A. (2010). MBTI personality type and other factors that relate to preference for online versus face-to-face instruction. *13*, 89–95.
John, O. P., Donahue, E. M., & Kentle, R. L. (1991). *The Big Five Inventory.* Berkeley: University of California, Berkeley.
Joksimovic, S., Gasevic, D., Kovanovic, V., Adesope, O., & Hatala, M. (2014). Psychological characteristics in cognitive presence of communities of inquiry: A linguistic analysis of online discussions. *Internet and Higher Education, 22*, 1–10.
Joo, Y. J., Lim, K. Y., & Kim, J. (2013). Locus of control, self-efficacy, task value as predictors of learning outcome in an online university context. *Computers & Education, 62*, 149–158.
Jordan, K. N., & Pennbaker, J. W. (2017). The exception or the rule: Using words to assess analytic thinking, Donald Trump, and the American Presidency. *Translational Issues in Psychological Science, 3*, 312–316.
Kauffman, H. (2015). A review of predictive factors of student success in and satisfaction with online learning. *Research in Learning Technology, 23*, 1–13.
Keller, H., & Karau, S. J. (2013). The importance of personality in students' perceptions of online learning experience. *Computers in Human Behavior, 29*, 2494–2500.
Kerr, M. S., Rynearson, K., & Kerr, M. (2006). Student characteristics for online learning success. *The Internet and Higher Education, 9*, 91–105.
Komarraju, M., Karau, S. J., Schmeck, R. R., & Avic, A. (2011). The big five personality traits, learning styles, and academic performance. *Personality and Individual Differences, 51*, 472–477.
Kovanović, V., Joksimović, S., Waters, Z., Gašević, D., Kitto, K., Hatala, M., Siemens, G., & Kovanović, V. (2016). *Proceedings of the Sixth International Conference on Learning Analytics and Knowledge (LAK16), April 25-29.* UK: Edinburgh Pages 15-24, ACM, New York, NY, USA.
Lederman, D. (2018). Conflicted views of technology: A survey of faculty attitudes. *Inside Higher Education.*. Retrieved from https://www.insidehighered.com/news/survey/conflicted-views-technology-survey-faculty-attitudes.
Means, B., Toyama, Y., Murphy, R., Bakia, M., & Jones, K. (2013). *Evaluation of evidence-based practices in online learning: A meta-analysis and review of online learning studies.* Washington, D.C.: U.S. Department of Education, Office of Planning, Evaluation, and Policy Development.
Online Education Trends Report (2018). Retrieved from https://www.bestcolleges.com/perspectives/annual-trends-in-online-education/.
Pennebaker, J. W., Boyd, R. L., Jordan, K., & Blackburn, K. (2015). *The development and psychometric properties of LIWC2015.* University of Texas at Austin.
Pennebaker, J. W., Chung, C. K., Frazee, J., Lavergne, G. M., & Beaver, D. I. (2014). When small words foretell academic success: The case of college admissions essays. *PLoS One, 9*(12), e115844.
Pennebaker, J. W., & King, L. A. (1999). Linguistic styles: Language as an individual differences. *Journal of Personality and Social Psychology, 77*, 1296–1312.
Pittinger, D. J. (1993). The utility of the Myer-Briggs Type Indicator. *Review of Educational Research, 63*, 467–486.
Poropat, A. E. (2009). A meta-analysis of the Five-Factor Model of personality and academic performance. *Psychological Bulletin, 135*, 322–338 in improving student outcomes. National Institute for Learning Outcomes Assessment.
Ransdell, S. (2013). Meaningful posts and online learning in blackboard across four cohorts of adult learners. *Computers in Human Behavior, 29*, 2730–2732.
Richardson, M., & Abraham, C. (2009). *Conscientiousness and achievement motivation predict performance*.
Richardson, M., Abraham, C., & Bond, R. (2012). Psychological correlates of university students' academic performance: A systematic review and meta-analysis. *Psychological Bulletin, 138*, 353–387.
Rothbart, M. K. (2007). Temperament, development, and personality. *Current Directions in Psychological Science, 16*, 207–212.
Saklofske, D. H., Austin, E. J., Mastoras, S. M., Beaton, L., & Osbourne, S. E. (2012). Relationships of personality, affect, emotional intelligence and coping with student stress and academic success: Different patterns of associations for stress and success. *Learning and Individual Differences, 22*, 251–257.
Shea, P., & Bidjerano, T. (2009). Cognitive presence and online learner engagement: A cluster analysis of the community of inquiry framework. *Journal of Computing in Higher Education, 21*, 199–217.
Shea, P., Bidjerano, T., & Vickers, J. (2016). Faculty attitudes toward online learning: Failures and successes. *SUNY Research Network*. Retrieved from http://www.sunyresearch.net/hplo/wp-content/uploads/2016/03/AERA-2016-w-new-analysis-3.pdf.
Soto, C. J., & John, O. P. (2009). *Ten facet scales for the Big Five Inventory: Convergence with NEO PI-R facets, self-peer agreement, and discriminant validity. 43*, 84–90.
Tausczik, Y. R., & Pennebaker, J. W. (2010). The psychological meaning of words: LIWC and computerized text-analysis methods. *Journal of Language and Social Psychology, 29*, 24–54.
Tlili, A., Esssalmi, F., Jemni, M., Kinshuk, & Chen, N. (2016). Role of personality in computer based learning. *Computers in Human Behavior, 64*, 805–811.
Top Ten (2019). What are the best personality types for online learning? Retrieved from https://www.top10onlinecolleges.org/faq/what-are-the-best-personality-types-for-online-learning/.
Tsai, C. C., Chuang, S. C., Liang, J. C., & Tsai, M. J. (2011). Self-efficacy in internet based learning environments: A literature review. *Educational Technology & Society, 14*, 222–240.
Vedel, A. (2014). The big five and tertiary academic performance: A systematic review and meta-analysis. *Personality and Individual Differences, 71*, 66–76.
Vedel, A., & Poropat, A. (2017). Personality and academic performance. In V. Zeigler-Hill, & T. Shackelford (Eds.). *Encyclopedia of personality and individual differences* (pp. 1–9). Springer Retrieved from https://www.researchgate.net/publication/313869097_Personality_and Academic Performance.
von Stumm, S., Hell, B., & Chamorro-Premuzic, T. (2011). The hungry mind: Intellectual curiosity is the third pillar of academic performance. *Perspectives on Psychological Science, 6*, 574–588.
Wang, A. Y., & Newlin, M. H. (2000). Characteristics of students who enroll in and succeed in psychology web-based classes. *Journal of Educational Psychology, 92*, 137–143.
Wang, A. Y., & Newlin, M. H. (2002). Predictors of web-students' performance: The role of self-efficacy and reasons for taking an online class. *Computers in Human Behavior, 18*, 153–163.
Wladis, C., & Samuels, J. (2016). Do online readiness surveys do what they claim? Validity, reliability, and subsequent student enrollment decisions. *Computers & Education, 98*, 39–56.
Yoo, J., & Kim, J. (2013). Can online discussion participation predict group project performance? Investigating the roles of linguistic features and participation patterns. *International Journal of Artificial Intelligence in Education, 24*, 8–32.
Yukselturk, E., & Bulut, S. (2007). Predictors for student success in an online course. *Educational Technology & Society, 10*, 71–83.
Zhao, Y., Lei, J., Yan, B., Lai, C., & Tan, H. S. (2005). What makes the difference? A practical analysis of research on the effectiveness of distance education. *Teachers College Record, 107*(8), 1836–1884.
Zhu, M., Herring, S. C., & Bonk, C. J. (2019). Exploring presence in online learning through three forms of computer-mediated discourse analysis. *Distance Education, 40*, 205–225.